

eg. in Python

```
</> Python
```

```
numbers = [1, 2, 3, 4]
```

```
squared = list(map(lambda x: x * x, numbers))
```

```
print(squared)
```

• The ~~result~~ result:

[1, 4, 9, 16] : Each number has multiplied itself.

25. What is Cloud Computing?

• Cloud computing means using computing resources such as servers, storage and software over the Internet of relying only on your own physical computer infrastructure.

3 main service models are:

- IaaS - Infrastructure as a Service: Provider computing infrastructure such as virtual servers and storage. eg. Amazon EC2

- PaaS - Platform as a Service: Provides a Platform for developing and running applications eg. Google App Engine

SaaS: Software as a Service: Provider ready-to-use software over the internet. Microsoft 365

26. What is a Version Control System?

A Version Control System (VCS) records changes made to files over time. It allows developers to track changes, work together and return to earlier versions when necessary.

Git is one of the most widely used version control systems

A commit is a saved record of changes made to a project.

A branch is a separate line of development that allows someone to work on changes without immediately affecting the main version of a project

27. What is a Jupyter Notebook is an interactive environment where users can combine code, explanations, data and visualisations in one document.

It is most commonly associated with Python, although it supports other programming languages

It is popular among data analysts and data scientists because they can write code, analyse data, create graphs and document their work in the same place.

28. What is Python?

Python is a high-level programming language known for its relatively simple syntax and wide range of application.

It is popular in Data Analytics and AI because it has many libraries designed for data processing, statistics, visualisation and machine learning.

4 common libraries:

Pandas: Used for data manipulation & analysis
Numpy: Used for numerical calculations and working with arrays

Matplotlib: Used to create charts and visualisation

Scikit-learn: Used for machine-learning tasks such as classification, regression and clustering

SECTION D: DATA ETHICS, GOVERNANCE & CAREERS

29. What is Data Privacy?

Data privacy refers to protecting people's personal information as collected, stored, used and shared.

It is important because personal information can be sensitive and misuse can cause financial, personal or other harm.

The GDPR (General Data Protection Regulation) is a European Union data law that establishes rules for processing personal data. It applies to organisations handling personal data of people in the EU in circumstances covered by the regulation.

30 What is Data Ethics

Data ethics refers to principles for using data and AI responsibly and fairly

3 examples:

1. Using personal information without proper permission. This can violate people's privacy.
2. Using AI to make unfair decisions: A biased system could unfairly disadvantage certain people
3. Selling or sharing sensitive information without appropriate consent: This can expose people to privacy and security risks

31. What is data Governance

Data governance is the system of rules, responsibilities and processes used to ensure that an organisation's data is managed properly

Important components:

- Data quality
- Data privacy
- Access control
- Data security
- Data standards
- Compliance

32. What is Bias in AI?

AI bias occurs when an AI system produces systematically unfair or inaccurate results for certain people or groups

One example is a recruitment algorithm that learns from historical hiring data that contains existing biases.

33. Data Analyst vs Data Scientist vs Data Engineer vs AI Engineer.

Role	Main responsibility	Common skills
Data Analyst	Analyses data and creates reports to support business decisions	Excel, SQL, Power BI, Python, Tableau
Data Scientist	Builds statistical and machine learning models to solve complex problems	Python, stats, SQL, Scikit-learn
Data Engineer	Builds and maintains systems that collect, process & store data	SQL, Python, database, cloud platforms, data pipelines
AI Engineer	Develops and integrates AI and machine learning systems into apps	Python, machine learning, APIs, cloud AI tools

34. What is Business intelligence?

Business intelligence is the process of using data, reports and visualisations to help organisations understand their performance and make informed decisions

2 popular BI tools are:

Power BI: Microsoft's BI platform used to connect to data, analyse it and create interactive reports and dashboards

Tableau: A data-visualisation & BI platform used to explore data and create interactive visualisations

Section E - Personal Reflection

35. Which concept did you find most interesting or surprising.

JSON and Python seem to have a similar way of coding for output. In my observation coding language does not differ all that much as SQL seems to have similar qualities.

36. Which concept was most difficult to understand. The concept I found most difficult was the difference data warehouse, data lakes and data marts. They seemed similar at first but with the examples on the websites it seems they all have different purposes

31. How do you see data and AI affect your future

Already with AI, it proved to be a massively helpful tool especially with normal every questions or even understanding my current career venture of Data Analytics.

I am yet to experience data same more but based on the case studies I have completed. It helps companies be able to identify and solve business problems.

So both have proven to be useful in different manner but essentially having the same goal of making life easier.

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